

Lecture-01

- i) OS definition
- ii) Computer system organization
- iii) " " operation
- iv) Interrupt (broad question)
- v) " handling
- vi) " drive I/O cycle

* Kernel to ram এ নেওয়া → boot strap program
↓
main memory

- * Operating system main component kernel.
- * device controller ও local buffer of controller
- * Ram temporary storage
- * Rom permanent storage
- * CPU কাজ করতে থাকবে input output signal পাঠাবে down হবে then আবার আগের কাজ করবে এভাবে শেষ হবে আবার চলবে
- * IOS at first idle আবার অন্ত্রা থেকে শুরু করতে request দেনে down হবে then আবার কাজ করতে হবে
- * Interrupt দুই প্রকার trap and exception
- * Interrupt Handling two type
 - i) Polling → কোন device CPU porjonto signal/request নিতে না
 - ii) Vectored → request টা CPU তে নেওয়া device এর

Lec-2

তারিখ: 15-06-2026

uname -n

↳ kernel version [architecture 32 bit or 64 bit]

running → 86, 64, 32 bit

uname -m

↳ server architecture

pwd [print working directory]

I) server setup

II) case sensitive

III) Present working directory

IV) folder name (home / find)

ls → list

↳ current folder list

I) basic linux command

ls -a

I) ফাইল / ফোল্ডার এর নাম (.) দিয়ে শুরু হলে hidden file এর দ্বারা।

mkdir - dir

↳ space দিয়ে হবে folder এর নামকরণ।

I) folder create command

II) make directory

mkdir dir1

mkdir a b c d e } multiple folder

ls

cd → change directory

cd dir1

↳ folder এর ভিতরে

cd dir2

mkdir dir3

cd dir3

cd ..

এক step পিছনে নিয়ে যাবে

cd [/home/student/Desktop/dir1/dir2/dir3]

↳ path (absolute bar)

1) আগের জায়গায় নিয়ে যাবে (home directory)

cd ~/Desktop/dir1/dir2



relative path

cd .

touch f1

1) empty file create করে

ls

ls f1

↳ f1

cat > f1 [ভিতরে কিছু লিখে দেয় file এর]

this is a test

test

cat f1

1) file এর ভিতর content show করতে

Ctrl d → back এ আসতে

cat f1 >> f2

1) append করা

2) 1st file content 2nd file এ change হয়

cat f1

cat f2

cat f1 f2 > f3

1) দুই file এর content merge then তারেক্টো file এ রাখ

2) একসাথে merge করে যার নাম আছে তার তিনিয় রাখ

cat f3

cp f3 f4

cp → copy

cat f4

1) cp এর পরে source তারপরে destination copy এর command f3 এর f4 করা

→ যদি f4 নাম folder কোন folder এ থাকত তাহলে তার path দিতে হবে।

different folder থাকলে pura path create করা

mv f4 file4] rename দিয়ে file4 নাম করা

1)

rm file4
ls

delete/remove করা কোন folder

rm * → all file remove [folder delete হবে না]

ls

folder remove

rm dir] empty folder ~~হবে~~ folder delete হবে না

Lecture-2

System Call : double, right call → any kind of message call passing

Storage Device Hierarchy:

storage ↑ small
↓ larger

time ↑ faster
↓ slower

register → faster

cache → memory for fast access
main memory তে যাওয়া

main memory (ram)

non-volatile memory

primary → permanent

Secondary

tertiary → মেমোরি সমগ্র খুলা যায়

Multiprocessor:

Dual core design:

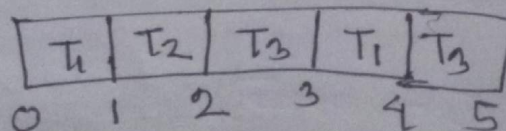
Clutered system:

SAN (Storage area network)

$$T_1 = 2^1$$

$$T_2 = 1$$

$$T_3 = 2^1$$



Dual mode / Dual mode system :

- I) User mode [mode bit = 1]
- II) kernel & (OS) [mode bit = 0]

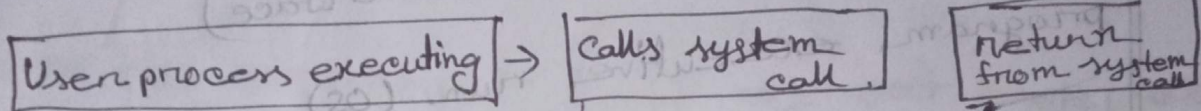
Lecture → 1, 2

wednesday

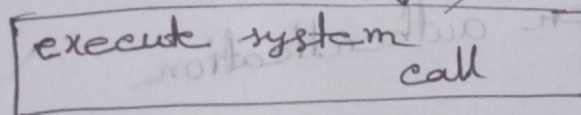
- I. Broad question
- II. Short & n (diagram must)

Transition from user to kernel Mode

User process



Kernel



Process Management:

program [main memory] → load → process

program → passive entity

process → active

caching:

ram → data copy → caching

Lecture 3

* Operating system services

- I) User interface (OS)
- II) terminal (CLI)
- III) GUI
- IV) touch screen (user interface)

program executive run (OS)

File S.M → OS features

protection → virus

security → user authentication

view of OSS:

diagram

System call:

API: Software to Software communication [Translator]

API

Windows → win 32

MAC / Linux → POSIX

Java → JVM

diagram

API → System call — OS Relationship

Parameter Passing via table

types of system calls

example of windows and unix system calls

Linux process creation relation fork

*** Linkers and Loaders



linker সব file কে merge/object করে একটি single file করে then Loader ham এ load করে execute

Implementation

Microkernel

MAC

cal → current month calendar

MCA
Lecture-3

cal - 3

↳ previous current next

ncal - M

↳ কারুর জন্য (S, M → Sunday, Monday ডায়েরী
অথবা বার হতে লা)

date

↳ current date time zone

date +%m

↳ month number

date +%h

↳ month name

দুইটি format একসাথে হলে

date +%h +%m

↳ month name + number

date +%d

↳ date টিউব

date +%M

↳ time → minute

date +%H

↳ hour

date +%r

↳ time with am/pm format

who

↳ server port [full information]

who am i

↳ connect student id (server)

logname

↳ login name

history

↳ all command show করতে (A to Z)

man date

↳ all date এর comment দেবে

Press quit দিলে বের

↳ press q

bc

↳ calculation → calculator

Ctrl - d → back

cat >f1

This is a test

test

wc f1

↳ word count

2 4 18 f1

↓ ↓ ↓
line word character

wc -c f1

↳ character

wc -l f1

↳ line

wc -w f1

↳ word

Lecture 4 (OBE)

OBE + Bread +
small question

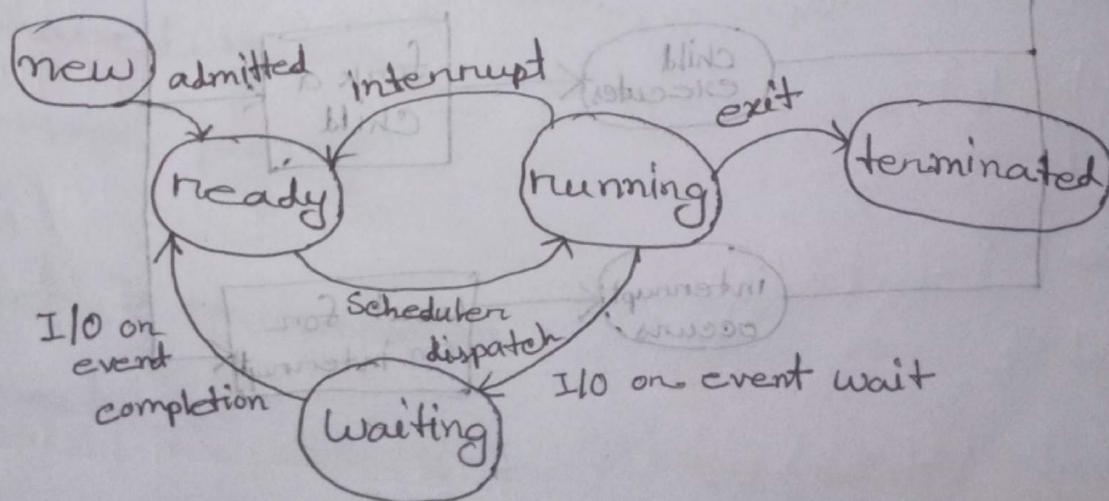
MCQ (lab)

Process State:

5 type

- I) new: create
- II) running: executed
- III) waiting: input and output
- IV) Ready: Ready for execution
- V) terminated: execution

Diagram

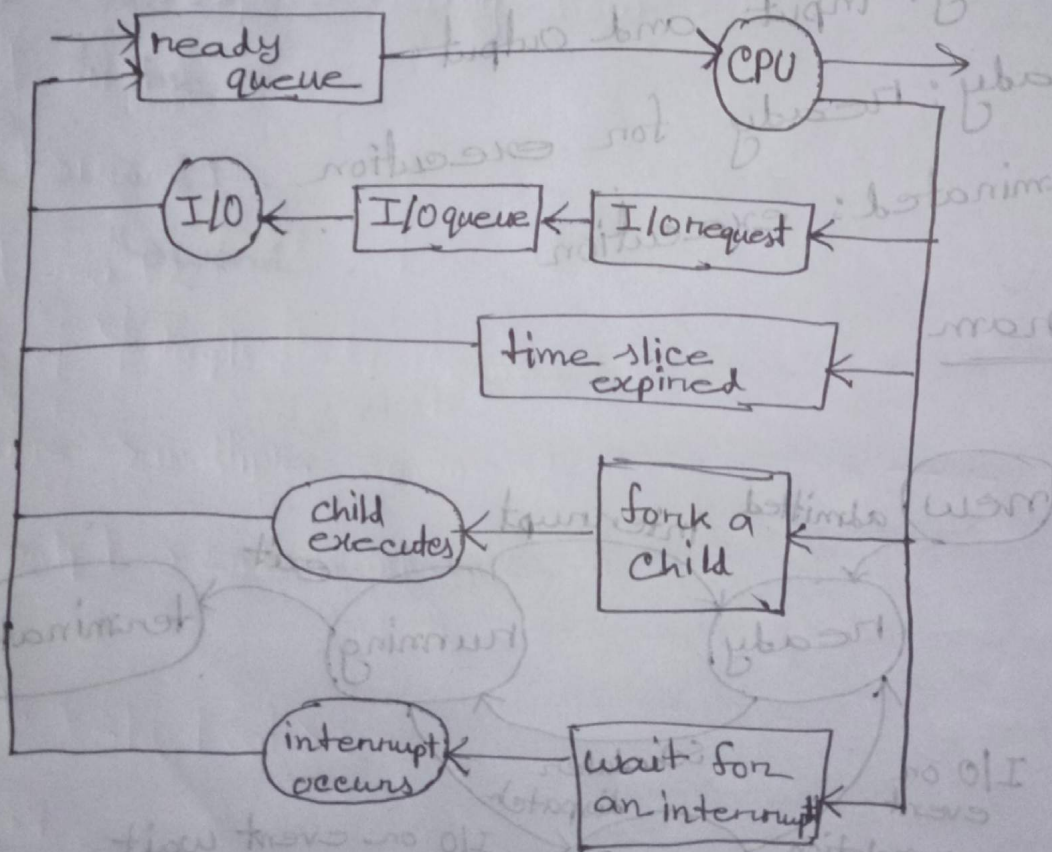


PCB (Process Control Block) :

Scheduling Queues :

- I) Job Queues : all process
- II) Ready " : ready set
- III) Device " : waiting set

Queuing diagram :



OBE
+
Broad
question

CPU switch from Process to Process:

Diagram
description

Schedulers:

short term: ready to select

Long term

context switch / CPU switch: same

Process creation:

Identify pid

Process termination:

Interprocess communication:

Shared memory
message passing

Direct communication:

Indirect

description [inter process
communication
সহ]

line number $\rightarrow$ nl f2

5 increment $\rightarrow$ i5 f2

head f2 $\rightarrow$ 1st 10 টি show করে

head -5 f2 $\rightarrow$ শেষ 5 টি " "

tail f2 $\rightarrow$ last 9 টি

tail -5 f2 $\rightarrow$ last 5 টি show করে

sort f2 $\rightarrow$ alphabetically show করে

sort -r f2

$\hookrightarrow$ reverse

sort -u f2

$\hookrightarrow$ duplicated remove

pipe line $\rightarrow$ multiple command একসাথে কাজ করে

1st output $\rightarrow$ 2nd command এর input

$\hookrightarrow$ |

sort f2 | nl | cat > 2

cat f4

sort f2 | cat > f2

cat f2

touch f2
cat > f2
file create

Lecture-6

multiple time - preemptive
একবার জেলে - non preemptive

Scheduling Criteria:

- i) CPU utilization (max)
- ii) Throughput (max)
- iii) Turnaround time (min)
- iv) Waiting time (min)
- v) response time (min)

FCFS

- i) show the gant chart
- ii) average waiting time = $\frac{\text{all}}{n}$
- iii) turn around time = waiting + burst time

SJF

non-preemptive

FCFS
SJF } non-preemptive

Shortest remaining time first: preemptive

Process	Arrival time	Brunt time
P ₁	0	8
P ₂	1	2
P ₃	2	9
P ₄	3	5

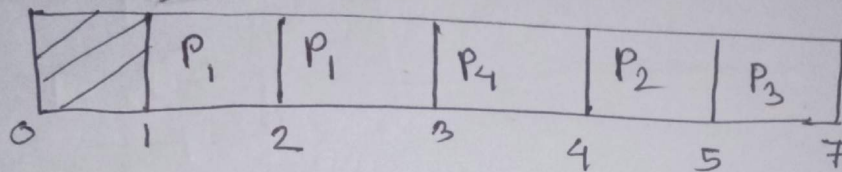
0 1 2 3 5 10 17 26

P₁ P₂ P₄ P₁ P₁ P₃

$$\text{Average time} = [(10-1) + (1-1) + (17-2) + (5-3)] / 4 = 110$$

$$\frac{110}{4} = 27.5$$

Process	Arrival Time	Burst Time
P_1	1ms	2ms
P_2	4ms	1ms
P_3	3ms	2ms
P_4	2ms	1ms



Waiting time

$$P_1 = (1-1) = 0$$

$$P_2 = (4-4) = 0$$

$$P_3 = (5-3) = 2$$

$$P_4 = (3-2) = 1$$

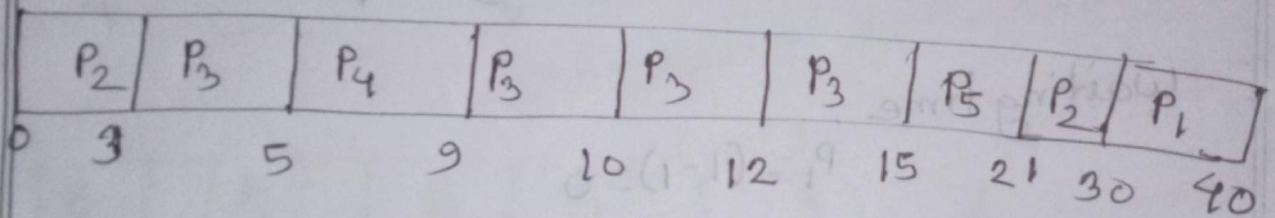
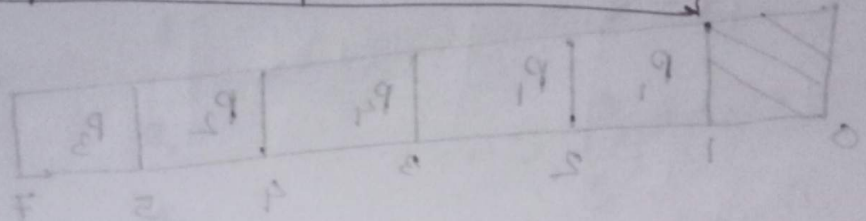
average time :

$$P = (2-1) + (4-4) = 1$$

$$0 = (2-2) = 0$$

$$2 = (5-3) = 2$$

Processes	Arrival time	Burst time
P ₂	0ms	12ms
P ₃	3ms	8ms
P ₄	5ms	4ms
P ₁	10ms	10ms
P ₅	12ms	6ms



waiting time,

$$P_1 = (30 - 10) = 20$$

$$P_2 = (0 - 0) + (21 - 3) = 18$$

$$P_3 = (3 - 3) + (9 - 5) = 4$$

$$P_4 = (5 - 5) = 0$$

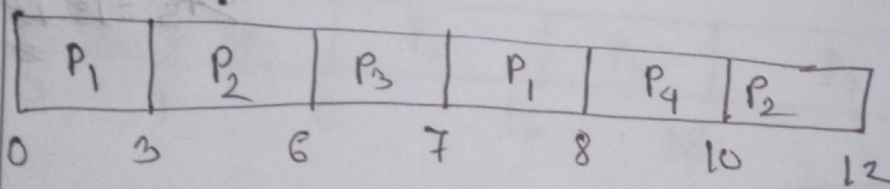
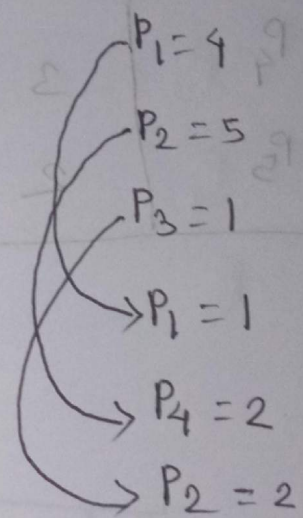
$$P_5 = (15 - 12) = 3$$

Round Robin

time quantum, $Q = 2$

Ready queue

Process	Arrival time	Burst time
P_1	0ms	4ms
P_2	1ms	5ms
P_3	2ms	1ms
P_4	5ms	2ms



waiting time

$$P_1 = 0 + (7 - 3) = 4$$

$$P_2 = (3 - 1) + (10 - 6) = 6$$

$$P_3 = (6 - 2) = 4$$

$$P_4 = (8 - 5) = 3$$

Q=3

Process	Arrival time	Burst time (millisecond)
P ₁	5	12
P ₂	0	4
P ₃	0	2
P ₄	3	3
P ₅	2	18

P₂ = 4

P₃ = 2

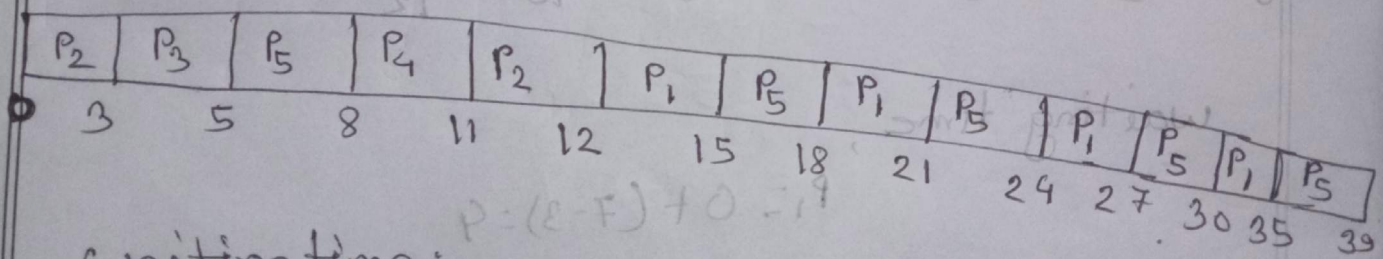
P₅ = 18

P₄ = 3

P₂ = 1

P₁ = 12

P₅ = 18



waiting time:

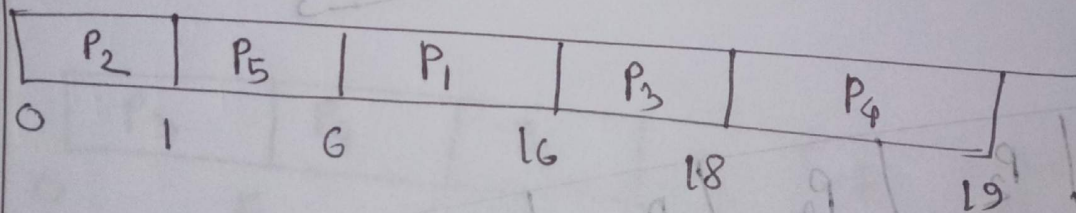
$W = (2 - 0) + (1 - 2) = 1$

$W = (3 - 0) = 3$

$W = (2 - 8) = -6$

Priority Scheduling:

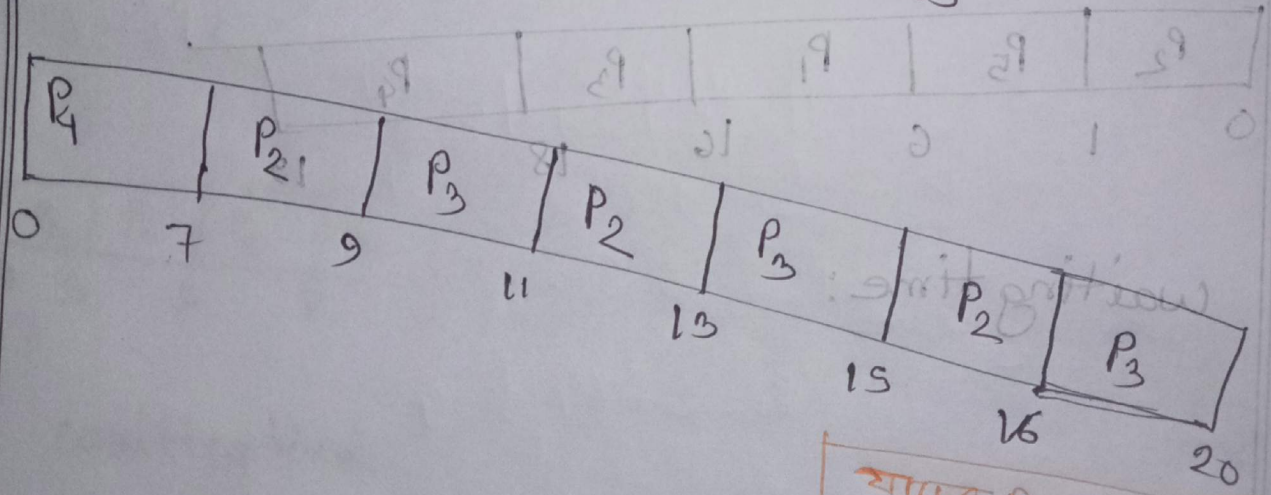
Process	Burst time	Priority
P ₁	10	3
P ₂	1	1
P ₃	2	4
P ₄	1	5
P ₅	5	2



waiting time:

Priority Scheduling w/ Round Robin :

Process	Burst time	Priority
P ₁	4	3
P ₂	5	2
P ₃	8	1
P ₄	7	2
P ₅	3	3



যাদের Priority double
তাদের round robin
নিজের মত

$Q_2 = 2$

Process	Burst time	Priority
P_1	3ms	5
P_2	5ms	3
P_3	6ms	1
P_4	1ms	2
P_5	4ms	3

